

BIOCHEMICAL STUDIES OF MARINE FISH OIL, PART II: EFFECTS OF FISH OIL AND LIPID LOWERING DRUGS ON TOTAL CHOLESTEROL AND TOTAL LIPID

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ABSTRACT

The fish oil alone and in combination with other antilipidemic drugs are studied in detail with special reference to cardiac ailments and this paper is in continuation of the same context carried out on fish oil as antilipidemic agents [1-3]. The naturally occurring ω -3 fatty acid (PUFAs), eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) in fish oil had led to a detailed investigation about its effect alone and in combination with competitive inhibitor HMG-CoA reductase (lovastatin) and it was found to be a potent lipid and cholesterol lowering agent. In the present study, the synergistic effects of fish oil with that of lovastatin or gemfibrozil were studied in order to reduce the quantity of these drugs to attain a better profile of activity thus avoiding side effects and toxicity of these drugs. As a result a better and workable formulation of a combined product was developed.

Keywords: Antilipidemic drugs, HMG-CoA reductase, Lovastatin, Gemfibrozil.

INTRODUCTION

This research work is in continuation of the same context carried out on fish oil as antilipidemic agents (Saify *et al.*, 2000, 2003a & b, 2004). It is significant that ω -3 PUFAs of fish oil rich in 20:5, 22:6 ω -3 appear to be much more effective in reducing hyperlipidemia than vegetable oils (Ponte, E. *et al.*, 1997 and Schmidt *et al.*, 1992). Seal and pilchords oil (Bronte-Stewart *et al.*, 1956), sardine oil (Anderson *et al.*, 1957), whale oil (Malmros and Wignad 1957) and freshly prepared menhaden oil (Ahrens *et al.*, 1959), have significant hypertriglyceridemic and hypercholesterole-mic effect in man. Significant reduction in plasma cholesterol and bile acids in aged rats than young indicate that the age enhance the effects of fish oil on plasma lipids by suppressing their characteristics fatty acid metabolism (Suzuki *et al.*, 1985). Nestel (1986) suggested that fish

oil rich in ω -3 fatty acids are capable of lowering lipoprotein cholesterol even when the make of cholesterol is high. Over the past quarter of a century, many studies have shown that large amounts of PUFAs in the diet exhibited significant hypolipidemic effect (Castro *et al.*, 1998, Van Vlijmen *et al.*, 1998). Feeding diet containing fish oil reduced the arachidonic acid content of rat serum and liver lipid fraction. The decrease being more pronounced when fish oil was fed in combination with hydrogenated beef tallow than with safflower oil. Apparently dietary ω -3 fatty acids of fish oil interact with dietary linoleic acid and saturated fatty acids differently to moderate enzymes of cholesterol and fatty acid metabolism. Elevated serum lipoproteins, prostaglandins and intracellular free calcium concentrations $[(Ca^{+2})]$ may be involved in the pathogenesis of cardiovascular diseases. Fish oil has been reported as having beneficial effects on these disorders (Locher *et al.*, 1989). Dagnelie *et al.* (1994) studied the effect of fish oil on blood free fatty acid,

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Table-1
Effect of Normal Diet on the Serum Total Cholesterol mg/100 ml in Group 1
Normal Control (N.C) Rabbits

WEEKS	1A	1B	1C	1D	1E	MEAN	±SE	P
0	59	34	35	35	38	40.20	4.25	
4	60	44	43	38	40	45.00	3.49	
8	65	68	48	46	62	57.80	4.04	
12	77	75	52	55	68	65.40	4.57	
16	80	80	61	65	80	73.20	3.77	
18	84	81	83	68	83	79.80	2.67	
20	93	96	87	78	87	88.20	2.76	
22	95	100	96	82	94	93.40	2.71	

Table-2
Effect of Cholesterol + Butter Rich Diet on the Serum Total Cholesterol mg/100 ml in
Group 2 Pathological Control (P.C) Rabbits

WEEKS	2A	2B	2C	2D	2E	MEAN	±SE	P
0	48	45	63	58	64	55.60	3.47	
4	238	198	263	245	190	226.80	12.57	
8	371	435	545	319	375	409.00	34.57	
12	529	565	714	551	495	570.80	33.72	
16	610	670	940	597	578	679.00	59.96	
Effect of normal diet*								
18	520	490	763	510	486	553.80	47.11	
20	322	376	595	381	437	422.20	41.93	
22	270	286	476	315	353	340.00	32.93	

*Normal diet: Loosan (alfa alfa), Gram seeds (chick pea)

Table-3
Effect of Cholesterol + Butter Rich Diet on the Serum Total Cholesterol mg/100 ml in
Group 3 Treated 1 (T.1) Rabbits

WEEKS	3A	3B	3C	3D	3E	MEAN	± SE	P
0	65	46	85	61	45	60.40	6.55	
4	275	205	247	235	255	243.40	10.38	
8	476	334	570	361	390	426.20	38.59	
12	537	458	641	509	437	516.40	32.07	
16	690	600	780	603	510	636.60	40.94	
Effect of fish oil (71.43 mg/kg/day)								
18	450	530	240	300	390	382.00	46.25	<0.01
20	395	370	68	220	373	285.20	55.99	<0.01
22	250	280	53	210	280	214.60	37.92	<0.01

ketone bodies and triacylglycerol in human and observed that blood triacylglycerol and free fatty acids decreased consistently.

Cholesterol in patients moved from blood and enters in the wall of aorta and other tissues (Biggs *et al.*, 1952). Lipoprotein as being a

filtrate of plasma deposited in the intima as foreign lipids. Importance of lipoprotein such as HDL, LDL, VLDL and chylomicron postulated by Fredrickson *et al.* (1967a, b) that HDL-apoprotein and LDL-apoprotein are found in α - and β -lipoprotein respectively. Both apoprotein are found in VLDL and

Table-4
Effect of Cholesterol + Butter Rich Diet on the Serum Total Cholesterol mg/100 ml in
Group 4 Treated 2 (T.2) Rabbits

WEEKS	4A	4B	4C	4D	4E	MEAN	±SE	P
0	15	15	24	28	22	20.80	2.29	
4	295	141	151	181	143	182.20	26.03	
8	478	169	176	270	181	254.80	52.56	
12	527	202	199	391	215	306.80	58.86	
16	667	229	218	484	244	368.40	80.01	
Effect of Lovastatin 10.57 mg/kg/day)								
18	367	137	93	363	168	225.60	52.01	NS
20	88	45	63	288	98	116.40	39.26	<0.05
22	37	27	51	250	56	84.20	37.36	<0.05

Table-5
Effect of Cholesterol + Butter Rich Diet on the Serum Total Cholesterol mg/100 ml in
Group 5 Treated 3 (T.3) Rabbits

WEEKS	5A	5B	5C	5D	5E	MEAN	± SE	P
0	21	20	27	16	22	21.20	1.58	
4	247	236	98	102	284	193.40	34.84	
8	356	387	133	143	436	291.00	57.04	
12	436	446	167	191	576	363.20	70.88	
16	519	558	199	212	696	436.80	88.48	
Effect of Gemfibrozil (17.14 mg/kg/day)								
18	281	288	126	183	402	256.00	42.52	NS
20	278	153	115	79	319	188.80	41.81	NS
22	243	67	87	56	295	149.60	44.44	<0.05

Table-6
Effect of Cholesterol + Butter Rich Diet on the Serum Total Cholesterol mg/100 ml in
Group 6 Treated 4 (7.4) Rabbits

WEEKS	6A	6B	6C	6D	6E	MEAN	± SE	P
0	69	56	42	38	48	50.60	4.93	
4	166	276	185	351	163	228.20	33.08	
8	410	395	242	420	281	349.60	32.83	
12	631	520	331	563	435	496.00	46.58	
16	870	610	470	630	580	632.00	58.68	
Effect of fish oil (71.43 mg/kg/day) + lovastatin (0.57 mg/kg/day)								
18	440	230	262	540	400	374.40	51.33	<0.01
20	364	157	95	380	245	248.20	50.03	<0.01
22	155	105	51	320	95	145.20	41.79	<0.01

chylomicron and takes part in transporting cholesterol in blood. Fish oil and fish may be useful as hypertriglyceridemia because of the presence of ω -3 PUFAs, EPA, DHA and it decreases VLDL, triglyceride synthesis.

Antihyperlipidemic drugs may be recommended when dietary and risk factor management fails, one of them is lovastatin that impairs the synthesis of cholesterol and VLDL by inhibiting HMG-CoA reductase.

Table-7
Effect of Cholesterol + Butter Rich Diet on the Serum Total Cholesterol mg/100 ml
in Group 7 Treated 5 (T.5) Rabbits

WEEKS	7A	7B	7C	7D	7E	MEAN	± SE	P
0	50	54	35	43	52	46.80	3.12	
4	355	340	353	155	263	293.20	34.41	
8	559	487	488	440	458	486.40	18.15	
12	635	601	537	753	608	626.80	31.67	
16	720	690	580	1170	685	769.00	92.13	
Effect of fish oil (71.43 mg/kg/day) + gemfibrozil (17.14 mg/kg/day)								
18	680	450	300	790	515	547.00	76.99	<0.05
20	396	397	145	352	382	334.40	42.97	<0.01
22	320	170	133	230	309	232.40	33.06	<0.01

Table-8
Effect of Cholesterol + Butter Rich Diet on the Serum Total Cholesterol mg/100 ml in
Group 8 Treated 6 (T.6) Rabbits

WEEKS	8A	8B	8C	8D	8E	MEAN	± SE	P
0	25	32	32	28	25	28.40	1.40	
4	336	217	287	95	90	205.00	44.43	
8	420	477	577	140	174	357.60	76.77	
12	537	503	745	215	276	455.20	85.50	
16	621	588	935	256	395	559.00	102.91	
Effect of lovastatin (0.57 mg/kg/day) + gemfibrozil (17.14 mg/kg/day)								
18	538	361	894	110	136	407.80	129.32	NS
20	532	220	337	80	87	251.20	75.75	NS
22	313	93	202	75	59	148.40	43.09	<0.05

Table-9
Effect of Cholesterol + Butter Rich Diet on the Serum Total Cholesterol mg/100 ml
in Group 9 Treated 7 (T.7) Rabbits

WEEKS	9A	9B	9C	9D	9E	MEAN	± SE	P
0	48	53	52	53	36	48.40	2.89	
4	411	291	285	252	235	294.80	27.58	
8	563	409	378	316	315	396.20	40.67	
12	711	560	515	435	486	541.40	42.05	
16	740	680	590	480	540	606.00	41.91	
Effect of fish oil (71.43 mg/kg/day) + lovastatin (0.57 mg/kg/day) + gemfibrozil (17.14 mg/kg/day)								
18	710	430	439	231	470	456.00	68.21	NS
20	532	268	298	62	400	312.00	69.51	<0.01
22	255	140	166	34	145	148.00	0.00	<0.01

Comparative studies of fish oil and lovastatin have been carried out with the aim to search out such a drug having high-potency and lesser side effects. The experiments were

carried out on rabbits and found synergistic effects of these compounds and such type of research is very valuable in curing modern world diseases.

Table-10
Effect of Normal Diet on the Serum Total Lipids mg/100 ml
in Group 1 Normal Control (N.C) Rabbits

WEEKS	1A	1B	1C	1D	1E	MEAN	±SE	P
0	200	195	180	200	209	196.80	4.27	
4	205	200	190	222	197	202.80	4.81	
8	230	205	200	231	225	218.20	5.85	
12	240	224	210	233	241	229.60	5.16	
16	250	225	200	247	256	235.60	9.24	
18	269	230	217	253	254	244.60	8.32	
20	293	255	231	267	271	263.40	9.09	
22	325	261	254	283	291	282.80	11.23	

Table-11
Effect of Cholesterol + Butter Rich Diet on the Serum Total Lipids mg/100 ml
in Group 2 Pathological Control (P.C) Rabbits

WEEKS	2A	2B	2C	2D	2E	MEAN	± SE	P
0	200	180	180	200	210	194.00	5.37	
4	379	315	390	345	381	362.00	12.54	
8	616	543	890	569	612	646.00	55.90	
12	763	681	1275	691	783	838.60	99.17	
16	956	850	1900	764	845	1063.00	189.14	
Effect of Normal diet*								
18	763	812	1206	634	755	834.00	87.21	
20	514	619	926	595	632	657.20	62.84	
22	456	597	673	454	560	548.00	37.67	

*Normal diet: Loosan (alfa alfa). Gram seeds (chick pea)

Table-12
Effect of Cholesterol + Butter Rich Diet on the Serum Total Lipids mg/100 ml
in Group 3 Treated 1 (7.1) Rabbits

WEEKS	3A	3B	3C	3D	3E	MEAN	± SE	P
0	170	190	185	185	180	182.00	3.03	
4	395	307	950	400	410	492.40	103.65	
8	615	591	1100	563	620	697.80	90.39	
12	856	687	1415	784	706	889.60	120.51	
16	960	920	2100	910	810	1140.00	215.80	
Effect of fish oil (71.43 mg/kg/day)								
18	800	813	1568	674	670	905.00	150.68	NS
20	780	780	1170	463	535	745.60	110.76	NS
22	450	450	701	410	480	498.20	46.43	<0.05

MATERIALS AND METHODS

- ☐ Electrical centrifuge 1000-300 rpm (China)
- ☐ Disposable syringe 10 ml, 21 gauge (B.P.)
- ☐ Centrifuge tube
- ☐ Glass tube
- ☐ Pasture pipettes
- ☐ Juster 0.1ml to 1 ml
- ☐ Microlab 100 (E. Merck)

Table-13
Effect of Cholesterol + Butter Rich Diet on the Serum Total Lipids mg/100 ml
in Group 4 Treated 2 (T.2) Rabbits

WEEKS	4A	4B	4C	4D	4E	MEAN	± SE	P
0	316	195	207	307	230	251.00	22.69	
4	550	390	507	527	563	507.40	27.62	
8	741	622	737	630	795	705.00	30.29	
12	901	790	805	800	888	836.80	21.26	
16	1504	893	952	970	925	1048.80	102.45	
Effect of Lovastatin (0.57 mg/kg/day)								
18	1363	378	833	815	776	833.00	140.21	NS
20	363	296	604	711	603	515.40	70.75	<0.01
22	336	215	250	601	225	325.40	64.50	<0.01

Table-14
Effect of Cholesterol + Butter Rich Diet on the Serum Total Lipids mg/100 ml
in Group 5 Treated 3 (T.3) Rabbits

WEEKS	5A	5B	5C	5D	5E	MEAN	± SE	P
0	268	243	230	263	356	272.00	19.76	
4	290	520	450	590	425	455.00	44.99	
8	388	640	524	818	716	617.20	66.88	
12	401	750	1060	990	868	813.80	103.74	
16	476	984	1552	1000	1831	1168.60	212.41	
Effect of Gemfibrozil (17.14 mg/kg/day)								
18	208	741	1272	811	1238	854.00	173.68	NS
20	145	605	901	483	992	625.20	135.92	<0.05
22	143	503	415	325	770	431.20	92.64	<0.05

Table-15
Effect of Cholesterol + Butter Rich Diet on the Serum Total Lipids mg/100 ml
in Group 6 Treated 4 (T.4) Rabbits

WEEKS	6A	6B	6C	6D	6E	MEAN	± SE	P
0	200	210	207	200	205	204.40	1.76	
4	311	389	380	583	270	386.60	48.14	
8	700	620	468	670	430	577.60	48.63	
12	816	795	560	835	583	717.80	53.82	
16	1000	920	615	970	730	717.80	66.75	
Effect of fish oil (71.43 mg/kg/day) + lovastatin (0.57 mg/kg/day)								
18	850	460	510	800	610	646.00	69.20	NS
20	780	381	210	670	310	470.20	97.41	<0.01
22	82	235	190	565	250	264.40	72.18	<0.01

Reagents:

All the reagent kits were purchased from Federal Republic of Germany.

Experimentally prepared food materials:

- Cholesterol (Crude, USA)
- Butter (Lurpak, Denmark)

Table-16
Effect of Cholesterol + Butter Rich Diet on the Serum Total Lipids mg/100 ml
in Group 7 Treated 5 (T.5) Rabbits

WEEKS	7A	7B	7C	7D	7E	MEAN	± SE	P
0	207	201	200	202	203	202.60	1.08	
4	500	515	491	280	363	429.80	41.39	
8	693	663	731	630	595	662.40	21.20	
12	887	814	845	856	816	843.60	12.13	
16	1100	986	910	930	894	964.00	33.44	
Effect of fish oil (71.43 mg/kg/day) + gemfibrozil (17.14 mg/kg/day)								
18	900	690	450	809	763	722.40	68.10	<0.01
20	650	600	290	610	650	560.00	61.06	<0.01
22	630	350	210	450	575	443.00	67.94	<0.01

Table-17
Effect of Cholesterol + Butter Rich Diet on the Serum Total Lipids mg/100 ml
in Group 8 Treated 6 (T.6) Rabbits

WEEKS	8A	8B	8C	8D	8E	MEAN	± SE	P
0	210	320	218	234	337	263.80	24.00	
4	215	490	302	880	770	531.40	115.39	
8	830	625	367	1531	1440	958.60	203.69	
12	950	1505	402	1590	1619	1213.20	211.58	
16	1844	1918	615	1676	1925	1595.60	222.91	
Effect of Lovastatin (0.57 mg/kg/day) + gemfibrozil (17.14 mg/kg/day)								
18	1088	1517	412	470	639	825.20	187.53	<0.05
20	599	1116	242	299	486	548.40	139.24	<0.01
22	320	456	175	286	367	320.80	41.42	<0.01

Table-18
Effect of Cholesterol + Butter Rich Diet on the Serum Total Lipids mg/100 ml
in Group 9 Treated 7 (T.7) Rabbits

WEEKS	9A	9B	9C	9D	9E	MEAN	± SE	P
0	210	180	210	170	230	200.00	9.80	
4	536	400	436	410	356	427.60	26.85	
8	656	636	562	560	576	598.00	17.92	
12	879	760	623	645	678	717.00	41.78	
16	960	910	690	740	810	822.00	45.20	
Effect of fish oil (71.43 mg/kg/day) + Lovastatin 10.57 mg/kg/day)								
18	885	680	565	460	680	654.00	63.33	NS
20	790	450	390	340	630	520.00	74.62	<0.01
22	410	210	278	282	220	280.00	31.88	<0.01

- ☐ Lovastatin
- ☐ Gemfibrozil
- ☐ Fish oil
- ☐ Chick pea (gram seed) local supplier
- ☐ Alfa alfa (loosan or green grass) local supplier

Animal and their feeding schedule:

A 22 weeks study involving 45 healthy, white male rabbits of 1.0 to 1.5 kg all of them were receiving their respective diet. The whole procedure from feeding schedule to the biochemical analysis and determination of total cholesterol and total lipid were explained comprehensively elsewhere in the thesis (Faheem Ahmed, 1995).

RESULTS

Fish oil obtained from two marine fish *Parastromateus niger* (Black Pomfret) and *Scomberomorus guttatus* (Surmai) (Faheem Ahmed, 1995) were examined alone and in combination with lovastatin and gemfibrozil. The results obtained on the parameters (tables 1-18) are given below:

1) Effects on total cholesterol:

Tables 1 and 2 represent the normal and pathological controls respectively. Significant effect on the plasma cholesterol level was observed in fish oil 66.29% ($P<0.01$, table-3), while the combination with lovastatin and with gemfibrozil exhibited 77.02% ($P<0.01$) and 69.78% ($P<0.01$) respectively (tables 6-7). However, fish oil + lovastatin + gemfibrozil showed a decrease of 75.58% ($P<0.01$, table-9). Lovastatin and gemfibrozil alone decreased the total cholesterol upto 77.14% ($P<0.05$) and 65.75% ($P<0.05$) respectively (tables 4-5) while the combination of these two drugs decreased the total cholesterol to 73.45% ($P<0.05$, Table-8).

2) Effects on total lipid:

Normal and Pathological control groups showing the effects on total lipid are given in tables 10 and 11 respectively. During the administration of fish oil, an overall decrease in total lipid was observed. However, maximum decrease was noted in case of lovastatin + gemfibrozil as 79.89% ($P<0.01$, Table-17) whereas in case of fish oil total lipid is decreased to 56.29% ($P<0.05$, table-12). The combined effect of fish oil + gemfibrozil + lovastatin exhibited a decrease of 65.93% ($P<0.01$, Table-18). Fish oil in combination

with gemfibrozil exhibited a decrease of 54.04% ($P<0.01$, Table-16) and fish oil + lovastatin showed a decrease of 68.78% ($P<0.01$, Table-15). It is interesting to observe that lovastatin showed a maximum decrease in lipid value as compared to that of gemfibrozil i.e., 68.97% ($P<0.01$) and 63.10% ($P<0.05$) respectively (Tables 13 and 14).

DISCUSSION

Our experimental results have shown specific effects of fish oil alone and in combination with other antilipidemic drugs to study a relationship between naturally occurring ω -3 fatty acid and drugs. Mostly we get the synergistic effects. Most significant effect produced in lowering the cholesterol level in case of fish oil and lovastatin may be attributed to the overall selective inhibition of HMG-CoA reductase by lovastatin (Tobert *et al.*, 1982; Brown and Goldstein, 1984). Thus the reduction in total cholesterol is achieved as a result of cumulative effect, which can further be expressed by the structure of ω -3 fatty acid in the same context. Further, reduction in plasma cholesterol levels can also be attributed to an increase excretion of sterols and bile acids in the feces, which have been observed before and after consumption of fish oil (Connor *et al.*, 1993). Also the mechanism of lovastatin as lipid lowering drug can be attributed to its interaction with enzyme HMG-CoA reductase, which may lead to a decrease in lipid content.

The observation of synergistic action of lovastatin and fish oil led to the conclusion that the site of action of fish oil may be the same as that of lovastatin (Huff *et al.*; 1992, Swahn and Olsson, 1993).

CONCLUSION AND FUTURE PROSPECTUS

During the course of these studies, we attempted to prove a better combination of two drugs namely gemfibrozil and lovastatin and also with fish oils. The effect of these two agents along with ω -3 fatty acid may lead to a

workable formulation of a combined product reducing the risk of side effects and toxicity. It can further be proposed that ω -3 fatty acid acts in synergistic action to that of lovastatin and this study would be helpful for the dieticians.

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